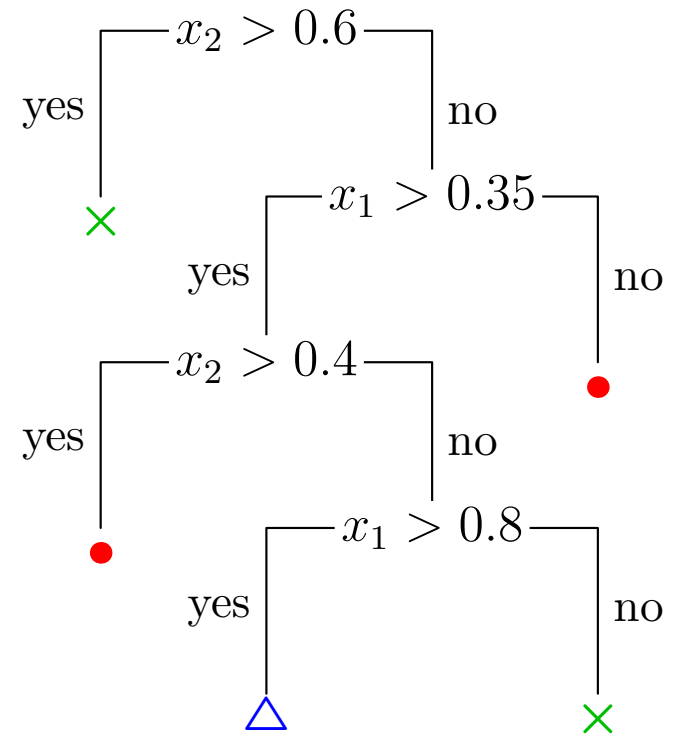
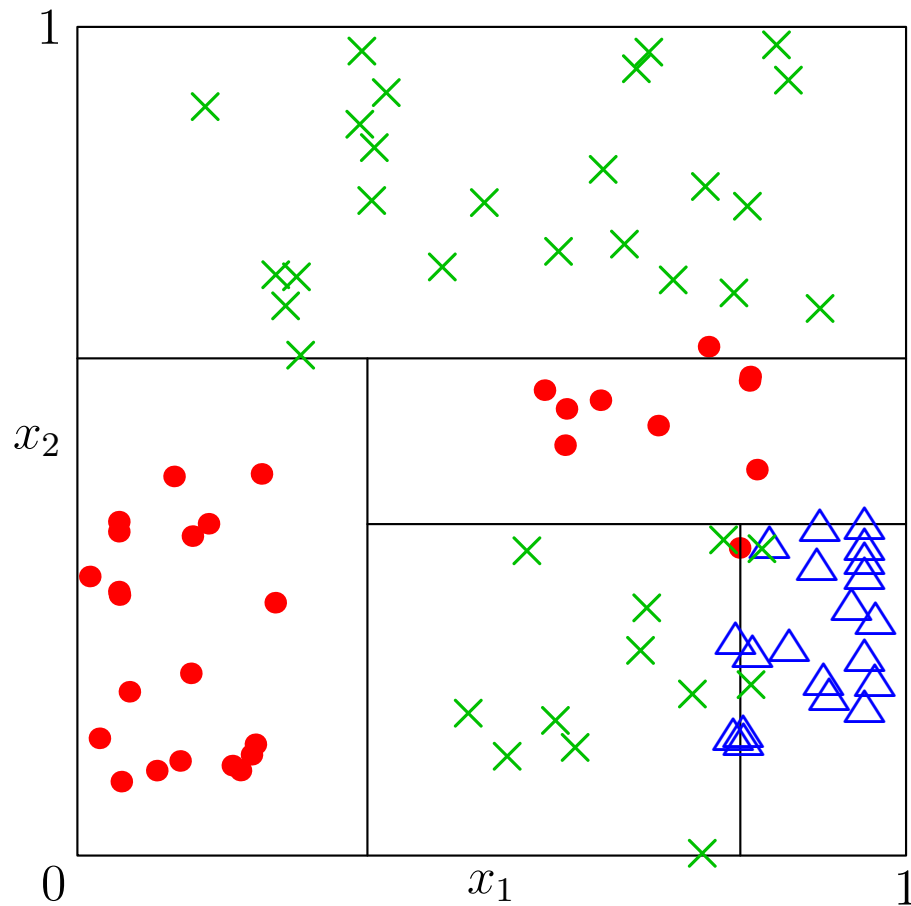


Advanced Machine Learning

Ensemble Methods

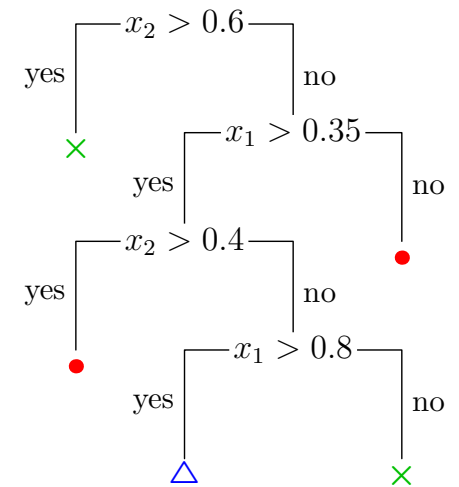
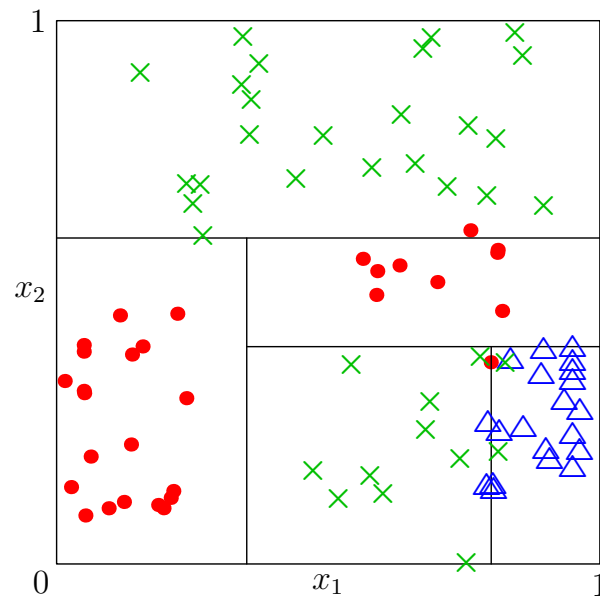


Decision Trees, Averaging, Bagging

Outline

1. Decision Trees

2. Bagging



Removing Variance By Averaging

- We can reduce the variance and hence improve our generalisation error by averaging over different learning machines
- There are a number of different techniques for doing this that go by the name of **ensemble methods** or **ensemble learning**
- This trick can be used with many different learning machines, but is clearly most practical for machine that can be trained quickly

Removing Variance By Averaging

- We can reduce the variance and hence improve our generalisation error by averaging over different learning machines
- There are a number of different techniques for doing this that go by the name of **ensemble methods** or **ensemble learning**
- This trick can be used with many different learning machines, but is clearly most practical for machine that can be trained quickly

Removing Variance By Averaging

- We can reduce the variance and hence improve our generalisation error by averaging over different learning machines
- There are a number of different techniques for doing this that go by the name of **ensemble methods** or **ensemble learning**
- This trick can be used with many different learning machines, but is clearly most practical for machine that can be trained quickly

Removing Variance By Averaging

- We can reduce the variance and hence improve our generalisation error by averaging over different learning machines
- There are a number of different techniques for doing this that go by the name of **ensemble methods** or **ensemble learning**
- This trick can be used with many different learning machines, but is clearly most practical for machine that can be trained quickly
- (nevertheless, even for deep learning taking the average response of many machines is usually done to win competitions)

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ **multiclass**
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Ensembling of Decision Trees

- One set of algorithms where ensembling are common place are decision trees
- These are particularly good for handling messy data
 - ★ categorical data
 - ★ mixture of data types
 - ★ missing data
 - ★ large data sets
 - ★ multiclass
- In many competitions ensembled trees, particularly *random forests* and *gradient boosting* beat all other techniques

Decision Trees

- A decision trees builds a binary tree to partition the data, $\mathcal{D} = \{(\mathbf{x}_i, y_i) | i = 1, \dots, m\}$, into the leaves of the tree
- Each decision rule depends on a single feature
- At each step the rule is chosen that maximise the “*purity*” of the leaf nodes
- Decisions can be made on numerical values or categories

Decision Trees

- A decision trees builds a binary tree to partition the data, $\mathcal{D} = \{(\mathbf{x}_i, y_i) | i = 1, \dots, m\}$, into the leaves of the tree
- Each decision rule depends on a single feature
- At each step the rule is chosen that maximise the “*purity*” of the leaf nodes
- Decisions can be made on numerical values or categories

Decision Trees

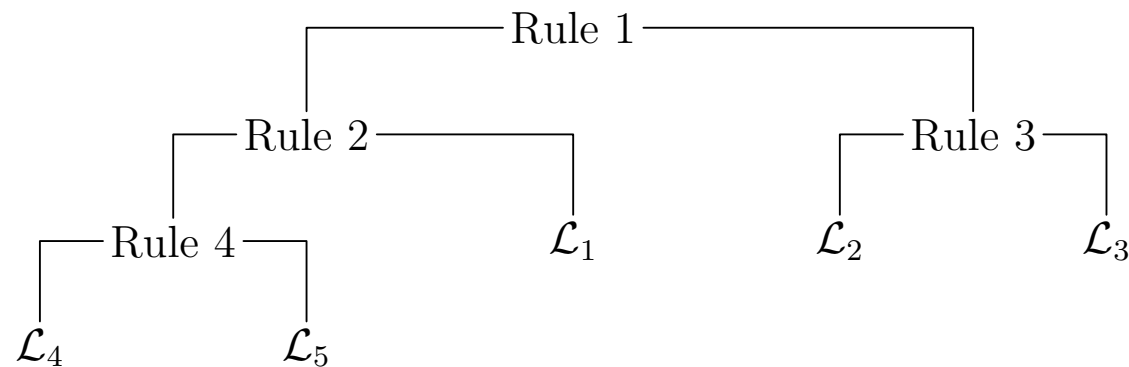
- A decision trees builds a binary tree to partition the data, $\mathcal{D} = \{(\mathbf{x}_i, y_i) | i = 1, \dots, m\}$, into the leaves of the tree
- Each decision rule depends on a single feature
- At each step the rule is chosen that maximise the “*purity*” of the leaf nodes
- Decisions can be made on numerical values or categories

Decision Trees

- A decision trees builds a binary tree to partition the data, $\mathcal{D} = \{(\mathbf{x}_i, y_i) | i = 1, \dots, m\}$, into the leaves of the tree
- Each decision rule depends on a single feature
- At each step the rule is chosen that maximise the “*purity*” of the leaf nodes
- Decisions can be made on numerical values or categories

Partitioning

- Consider a classification problems with examples (x, y) belonging to some classes $y \in \mathcal{C}$
- The data is partitioned by the tree into leaves



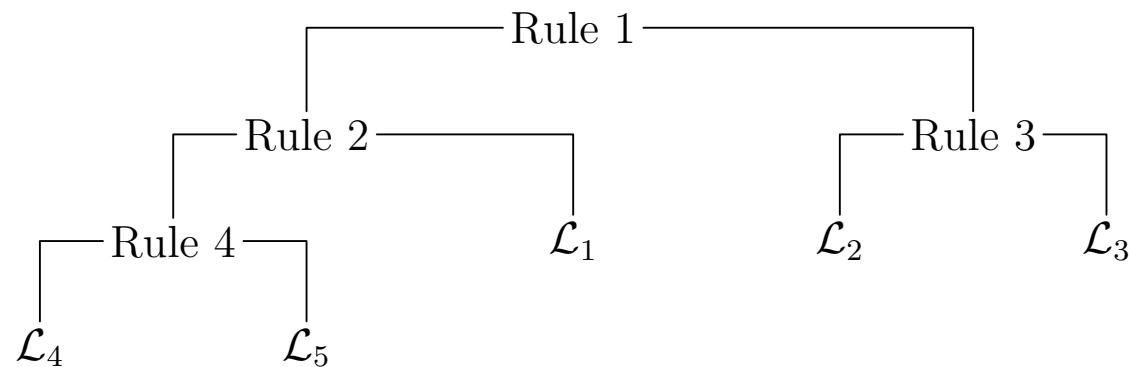
- The proportion of data points in leaf $\mathcal{L}$ belonging to class c is

$$p_c(\mathcal{L}) = \frac{1}{|\mathcal{L}|} \sum_{(x, y) \in \mathcal{L}} \mathbb{I}[y = c]$$

where $\mathbb{I}[y = c] = 1$ if $y = c$ and 0 otherwise

Partitioning

- Consider a classification problems with examples $(\mathbf{x}, y)$ belonging to some classes $y \in \mathcal{C}$
- The data is partitioned by the tree into leaves



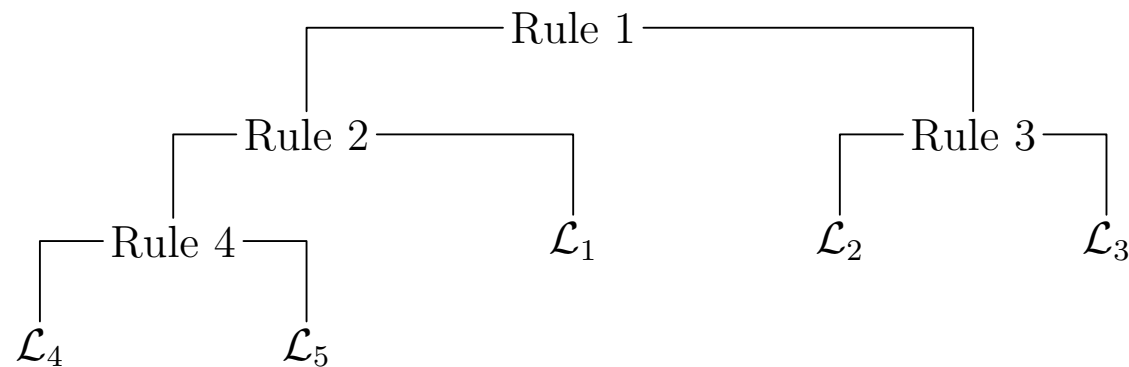
- The proportion of data points in leaf $\mathcal{L}$ belonging to class c is

$$p_c(\mathcal{L}) = \frac{1}{|\mathcal{L}|} \sum_{(\mathbf{x}, y) \in \mathcal{L}} \mathbb{I}[y = c]$$

where $\mathbb{I}[y = c] = 1$ if $y = c$ and 0 otherwise

Partitioning

- Consider a classification problems with examples $(\mathbf{x}, y)$ belonging to some classes $y \in \mathcal{C}$
- The data is partitioned by the tree into leaves



- The proportion of data points in leaf $\mathcal{L}$ belonging to class c is

$$p_c(\mathcal{L}) = \frac{1}{|\mathcal{L}|} \sum_{(\mathbf{x}, y) \in \mathcal{L}} \mathbb{I}[y = c]$$

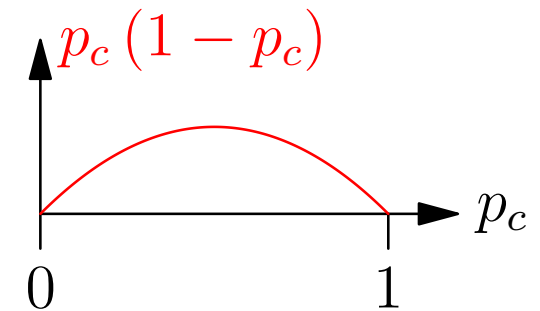
where $\mathbb{I}[y = c] = 1$ if $y = c$ and 0 otherwise

Leaf Purity

- Two different purity measures, $Q_m(\mathcal{L})$, for a leaf node $\mathcal{L}$ are commonly used

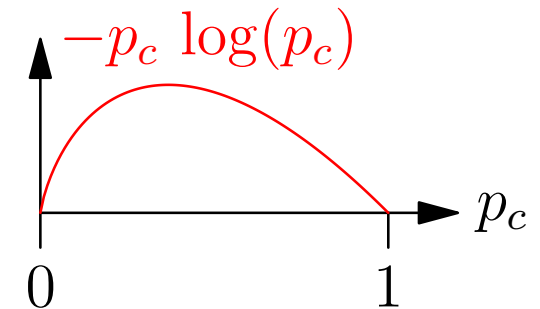
★ Gini index

$$Q_m^g(\mathcal{L}) = \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) (1 - p_c(\mathcal{L}))$$



★ Cross-entropy

$$Q_m^e(\mathcal{L}) = - \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) \log(p_c(\mathcal{L}))$$

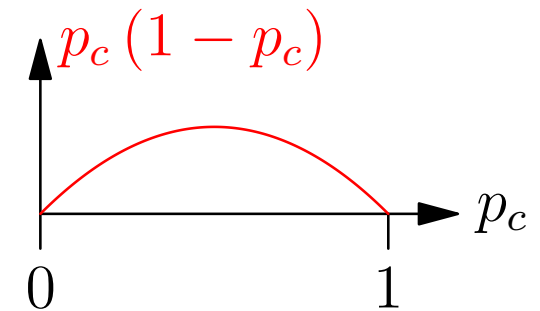


Leaf Purity

- Two different purity measures, $Q_m(\mathcal{L})$, for a leaf node $\mathcal{L}$ are commonly used

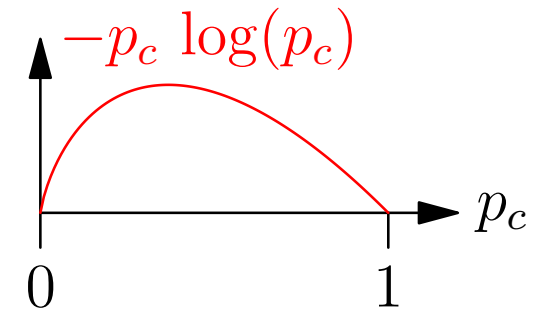
- ★ **Gini index**

$$Q_m^g(\mathcal{L}) = \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) (1 - p_c(\mathcal{L}))$$



- ★ **Cross-entropy**

$$Q_m^e(\mathcal{L}) = - \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) \log(p_c(\mathcal{L}))$$

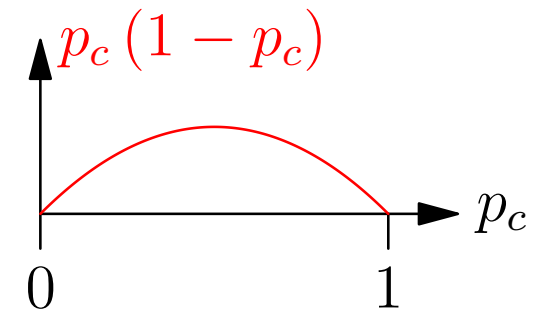


Leaf Purity

- Two different purity measures, $Q_m(\mathcal{L})$, for a leaf node $\mathcal{L}$ are commonly used

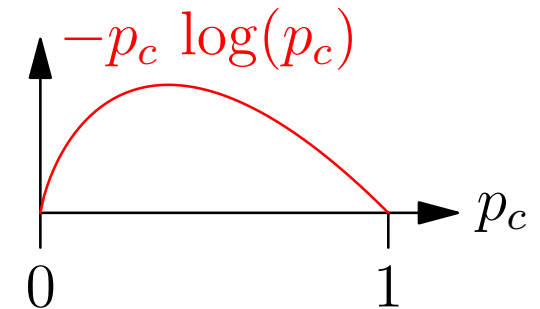
★ Gini index

$$Q_m^g(\mathcal{L}) = \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) (1 - p_c(\mathcal{L}))$$

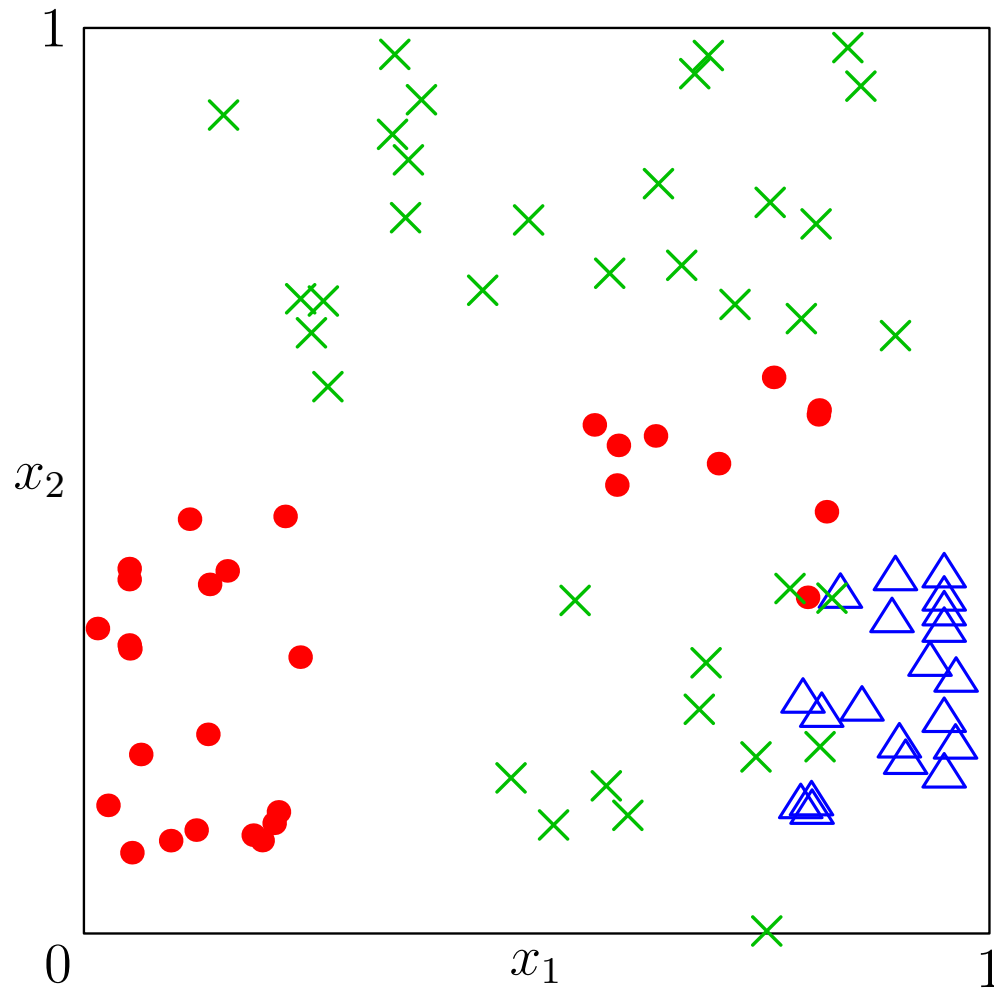


★ Cross-entropy

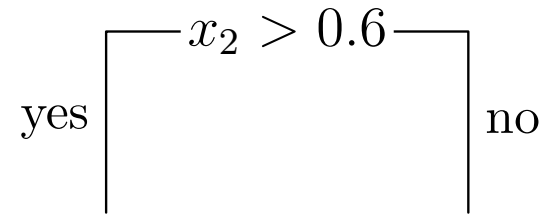
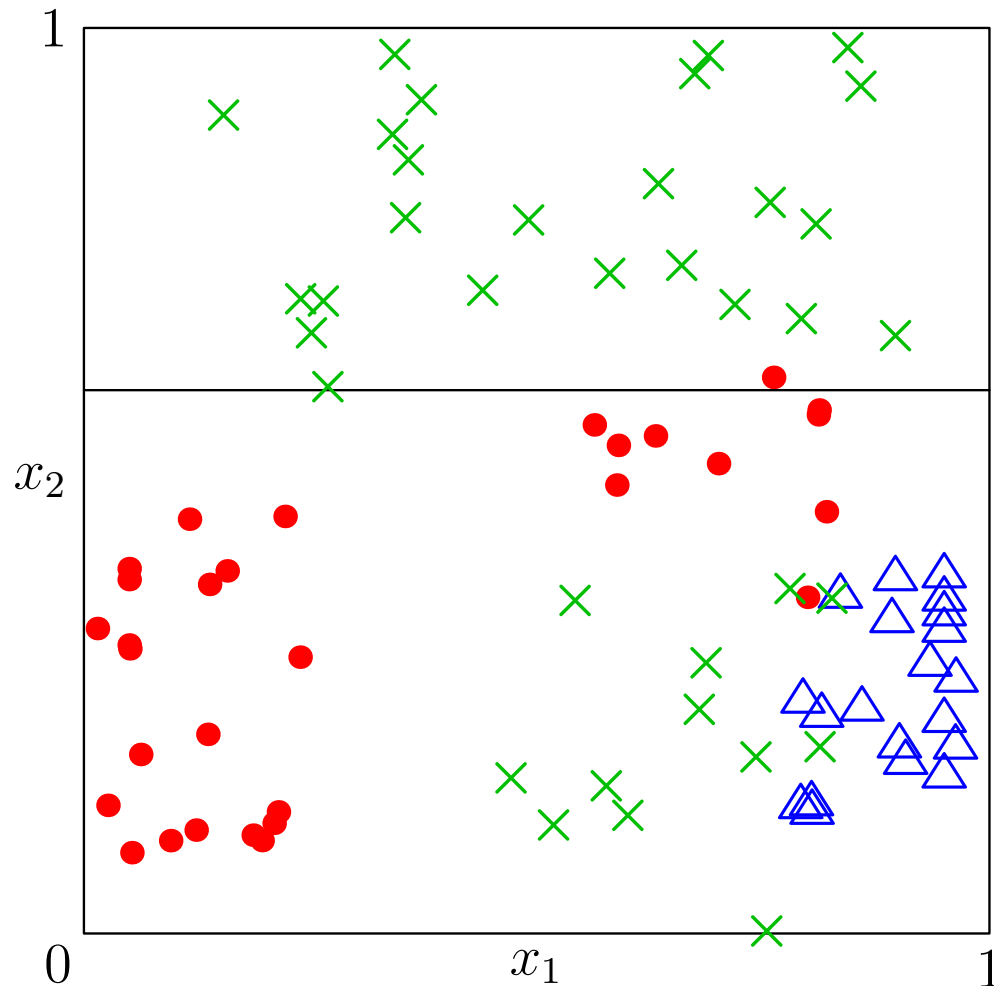
$$Q_m^e(\mathcal{L}) = - \sum_{c \in \mathcal{C}} p_c(\mathcal{L}) \log(p_c(\mathcal{L}))$$



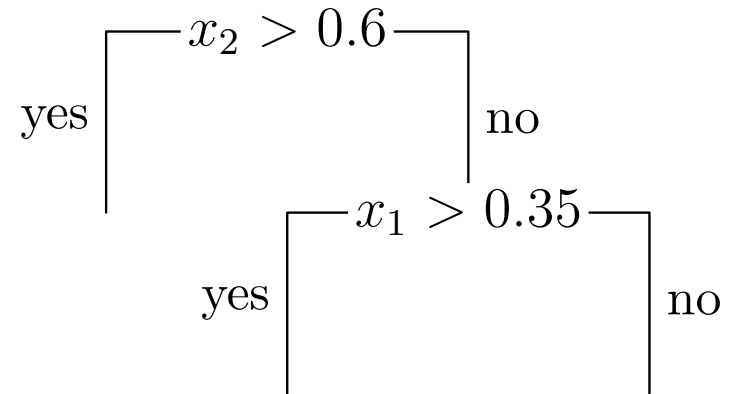
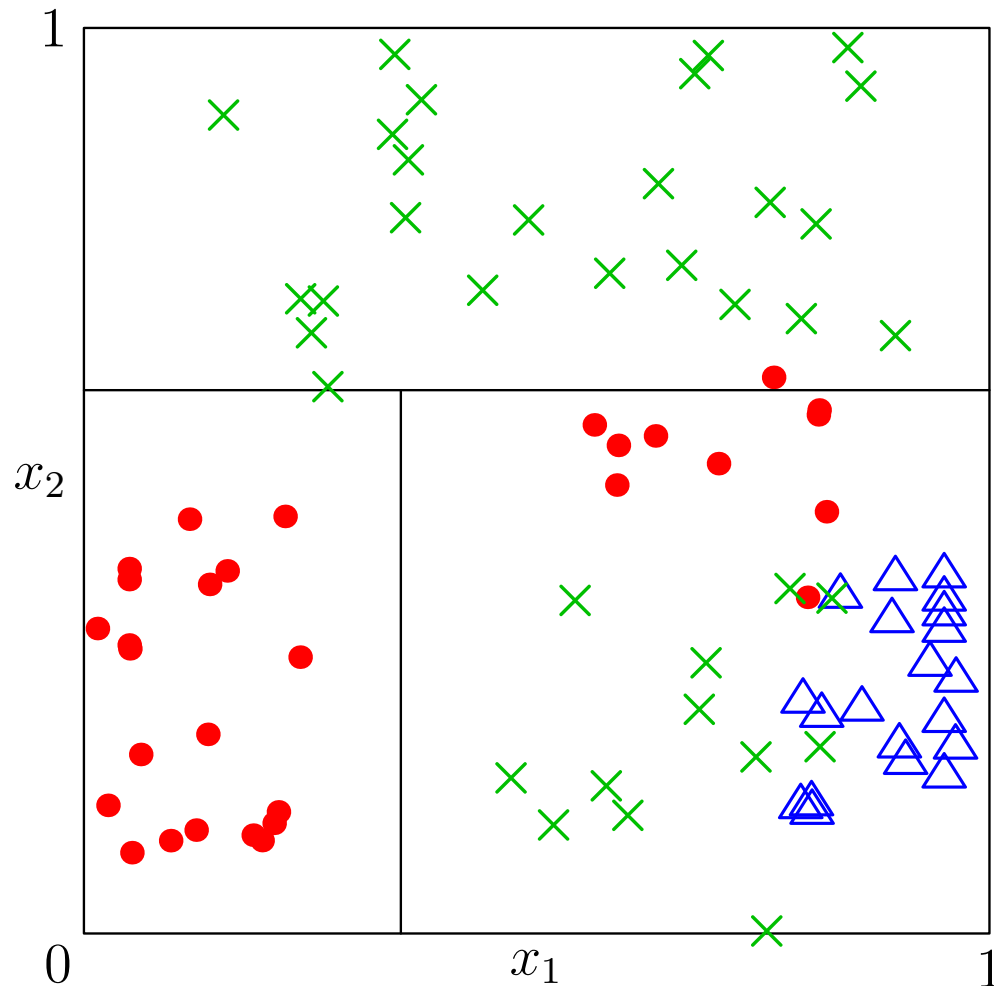
Building Decision Trees



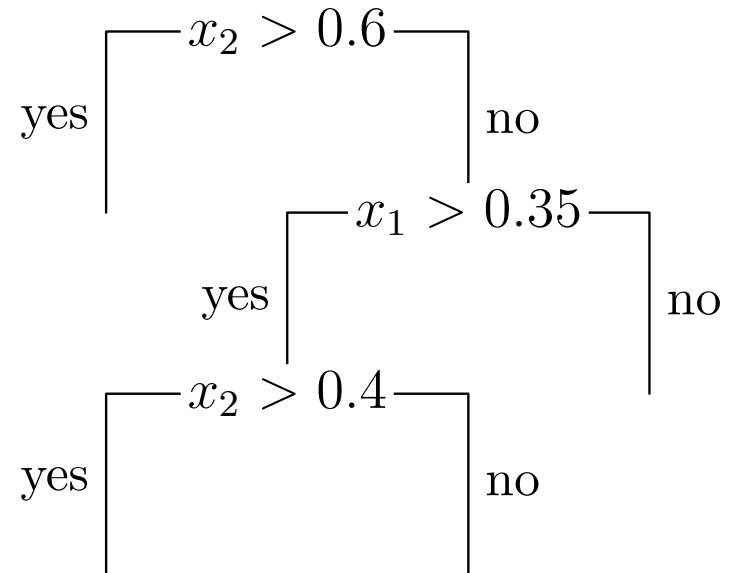
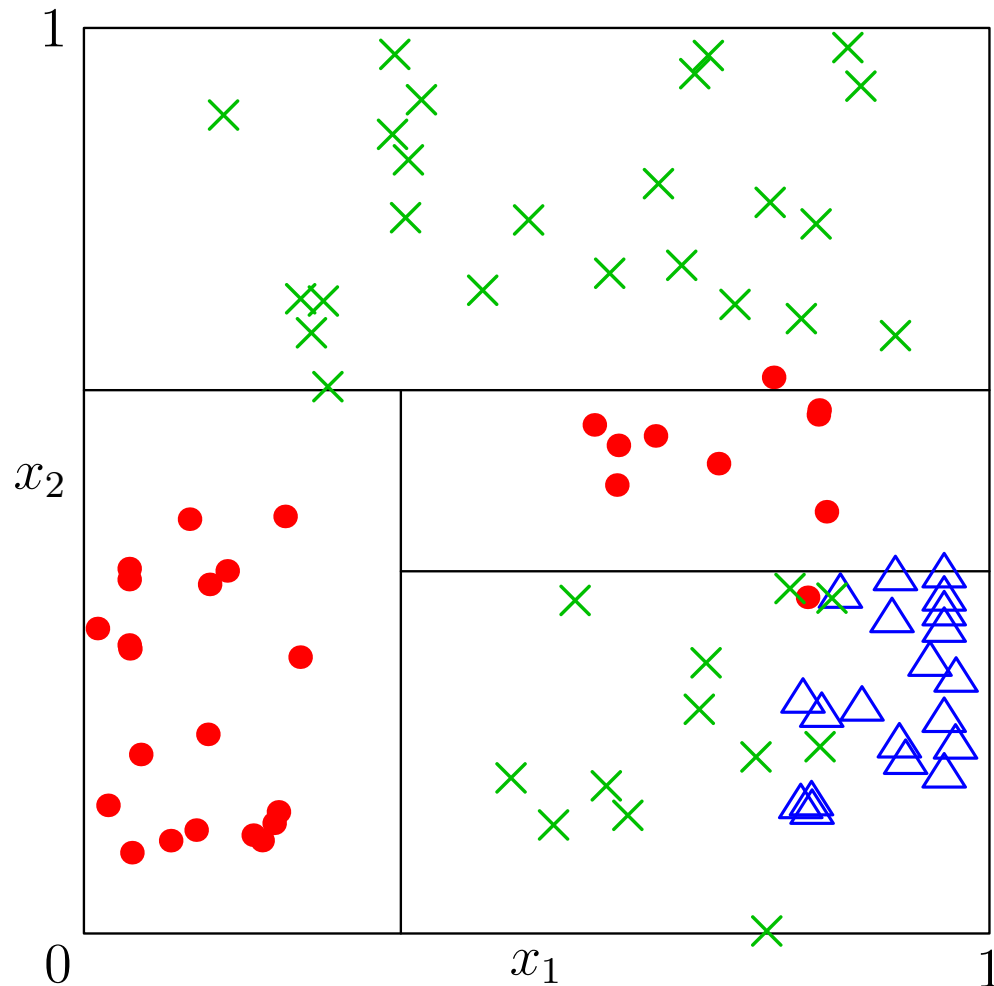
Building Decision Trees



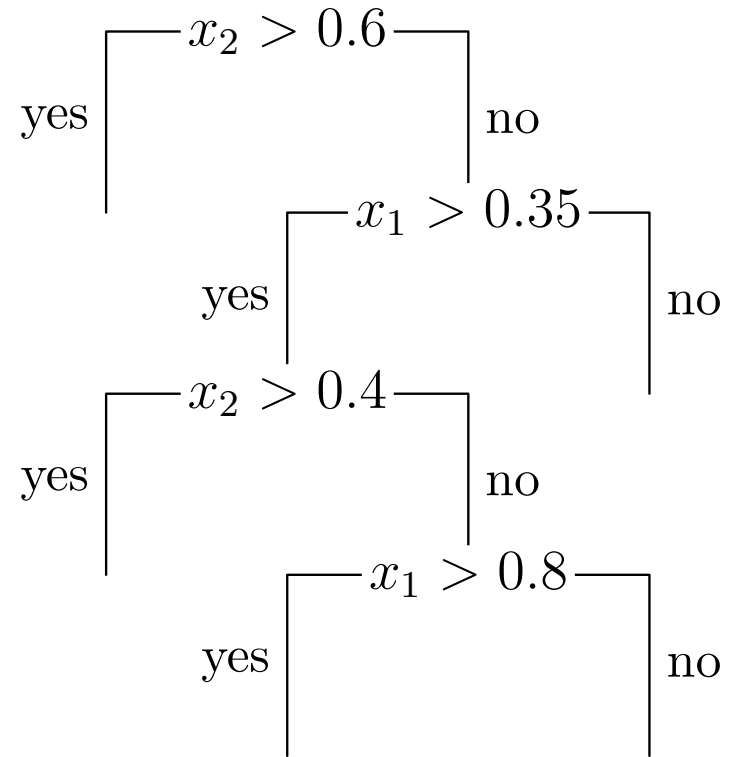
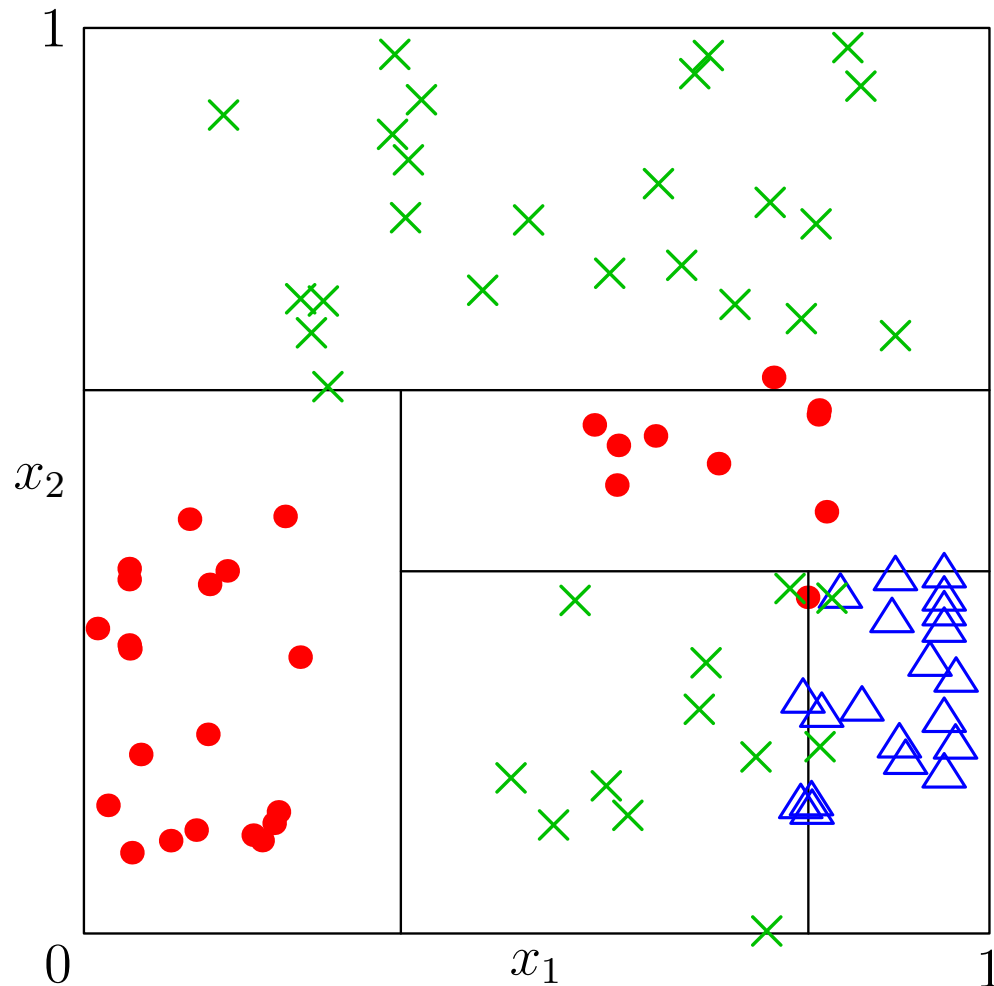
Building Decision Trees



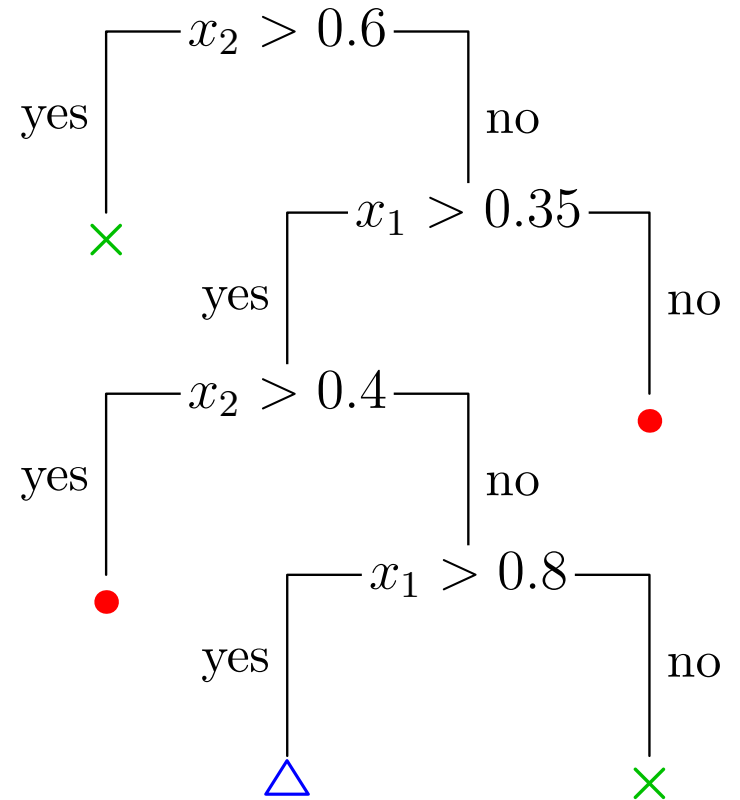
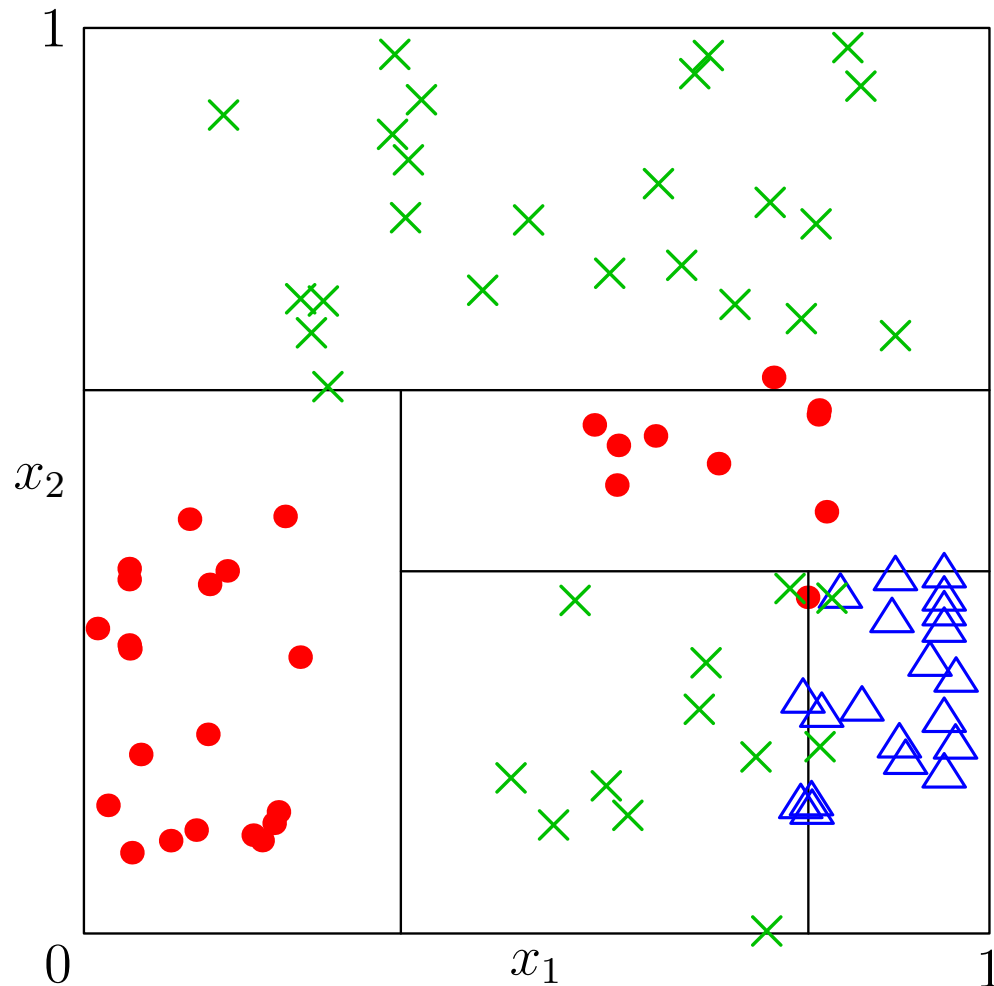
Building Decision Trees



Building Decision Trees



Building Decision Trees



Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds Classification And Regression Trees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets

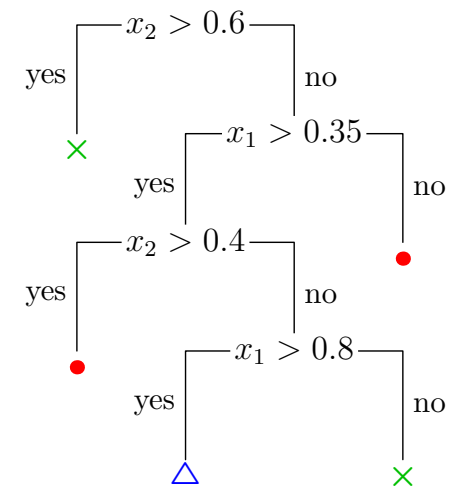
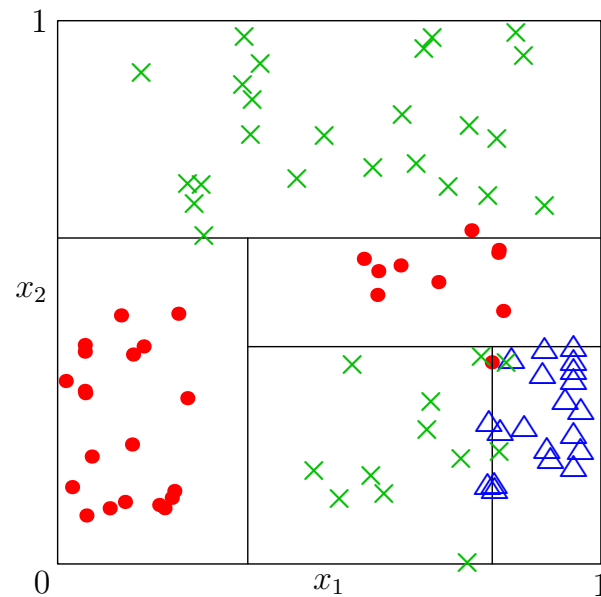
Observations

- Decision trees are very useful for exploring new data sets—the tree shows what features are most important
- Decision trees can also be used for regression problems
 - ★ Approximate function by a series of rules
 - ★ Reduce variance between data points assigned to leaf nodes
- CART is a classic implementation that builds **C**lassification **A**nd **R**egression **T**rees
- Decision trees depend strongly on the early decisions and so vary a lot for slightly different data sets—**high variance**

Outline

1. Decision Trees

2. Bagging



Error In The Means

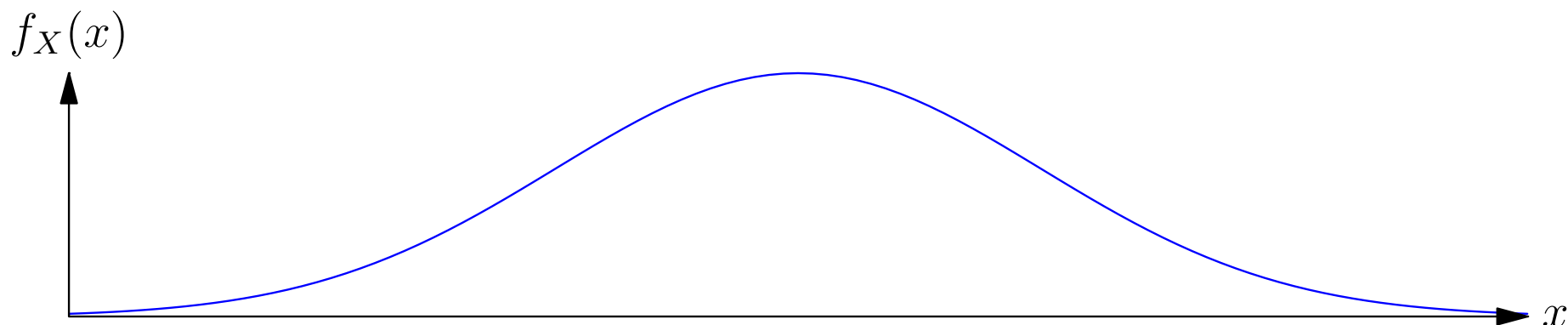
- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)

Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)

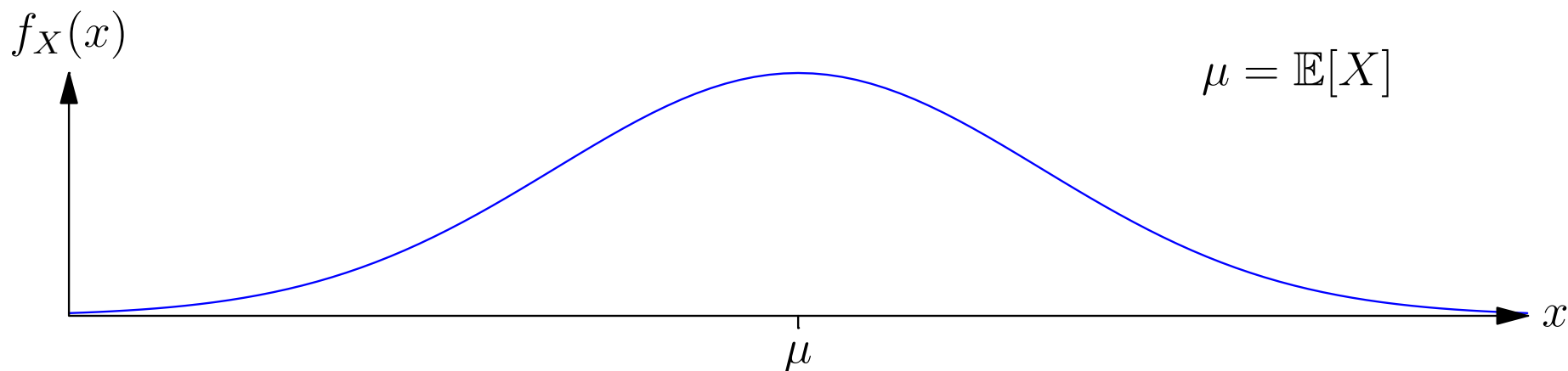
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



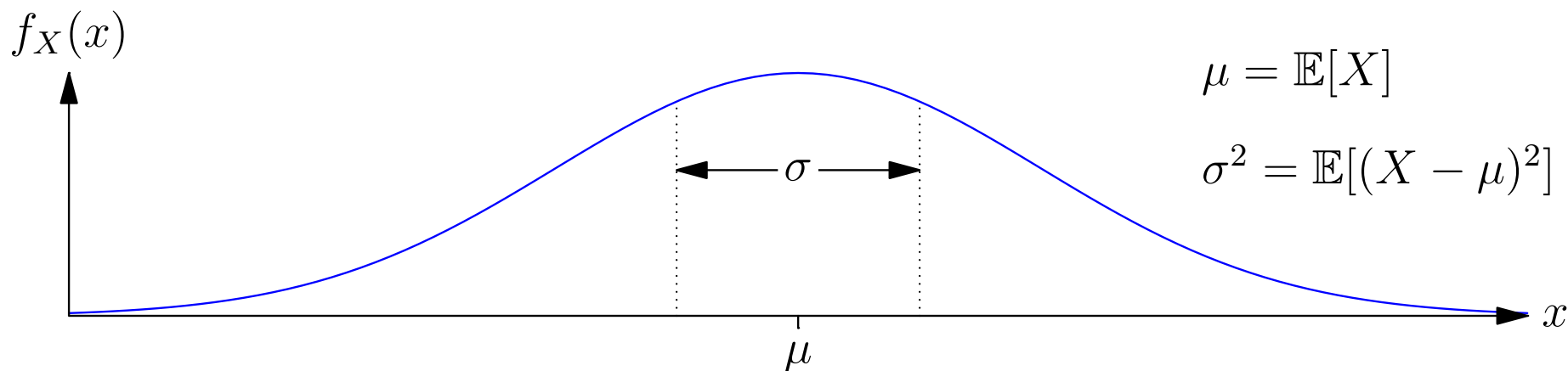
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



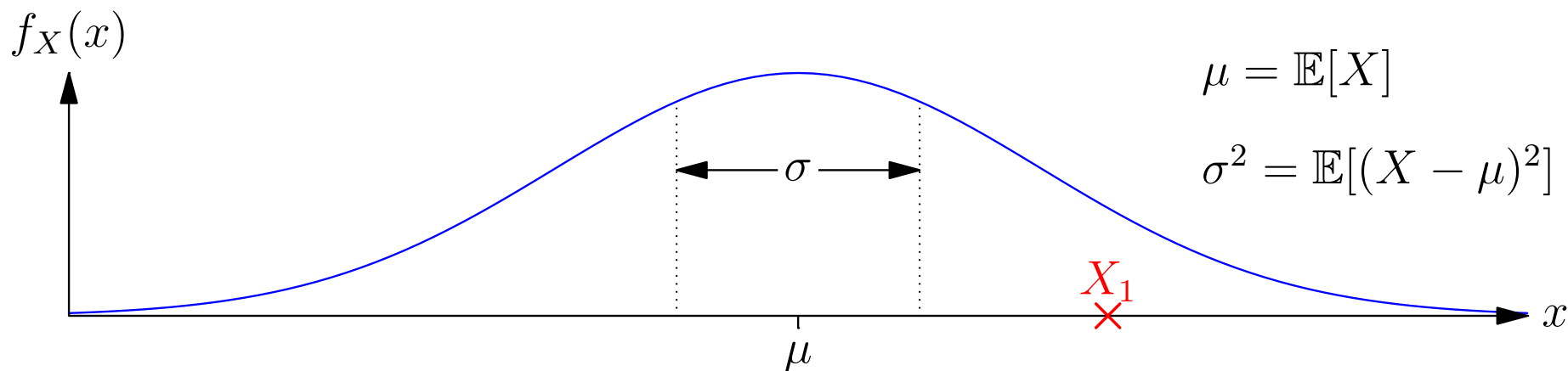
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



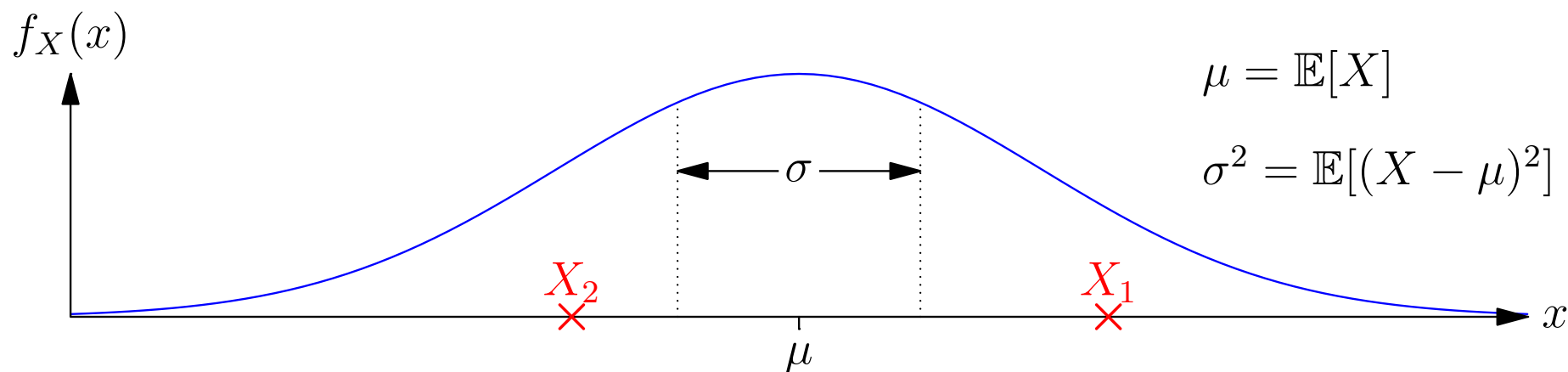
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



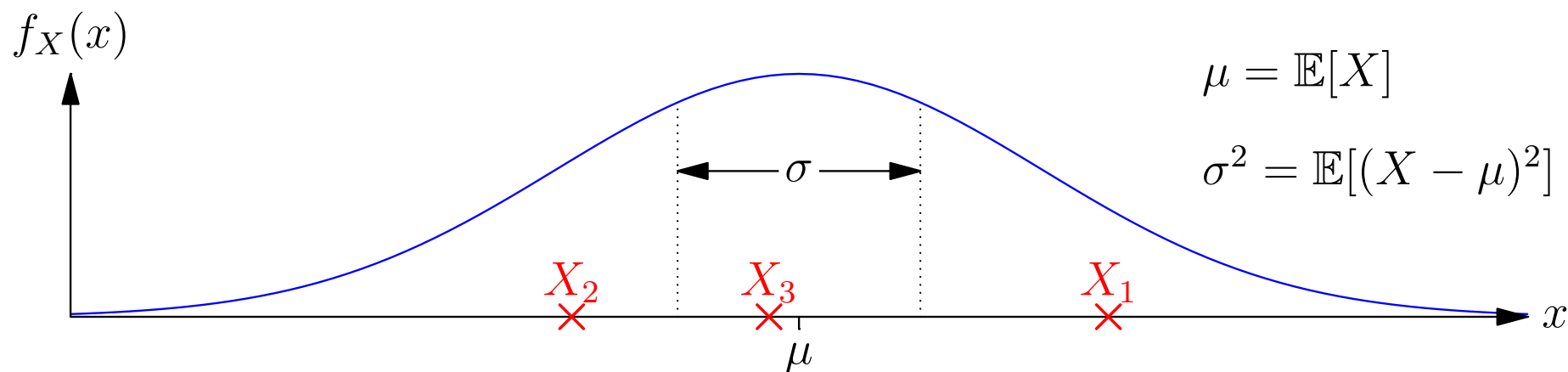
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



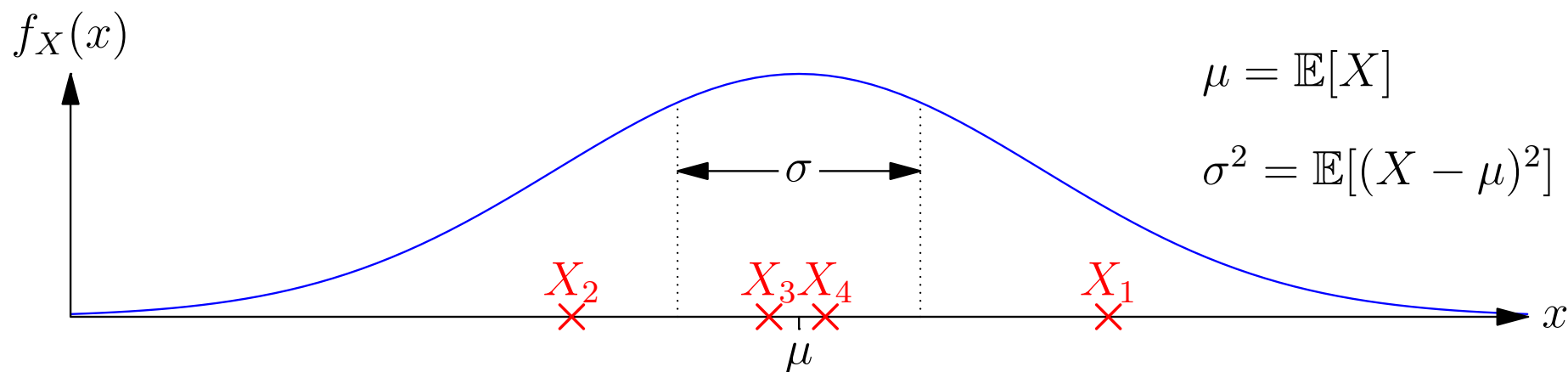
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



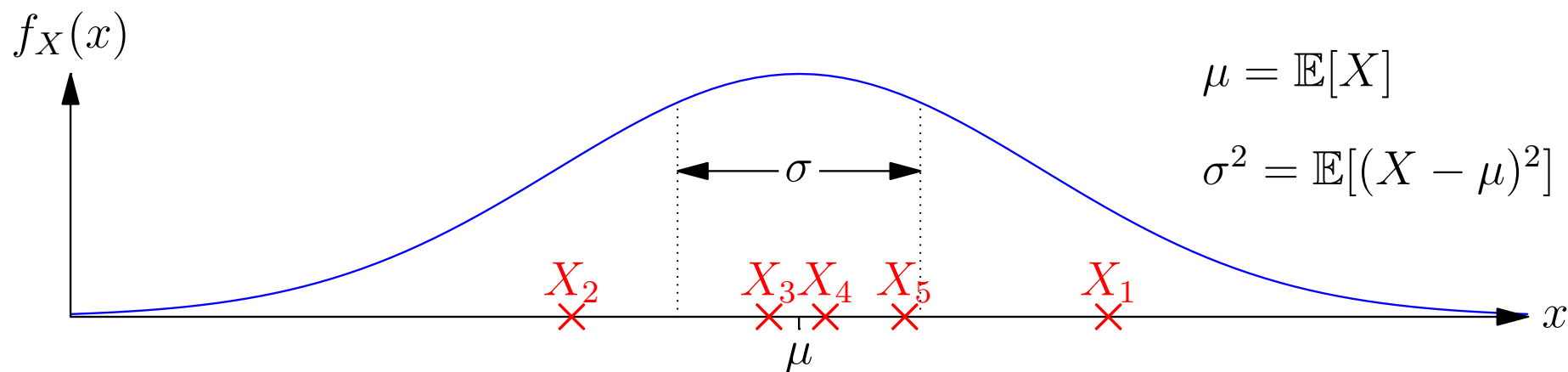
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



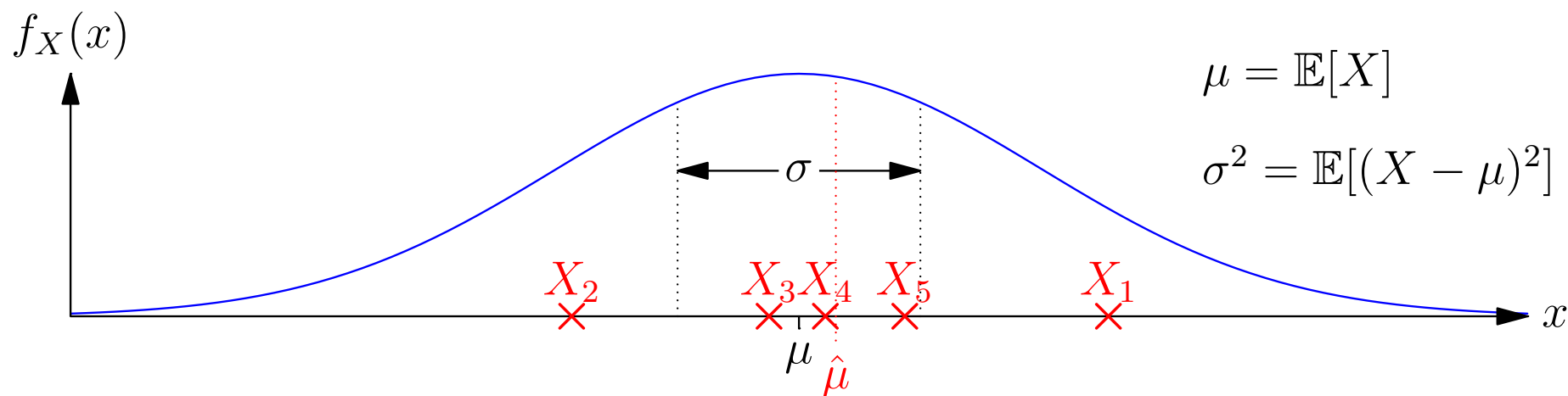
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



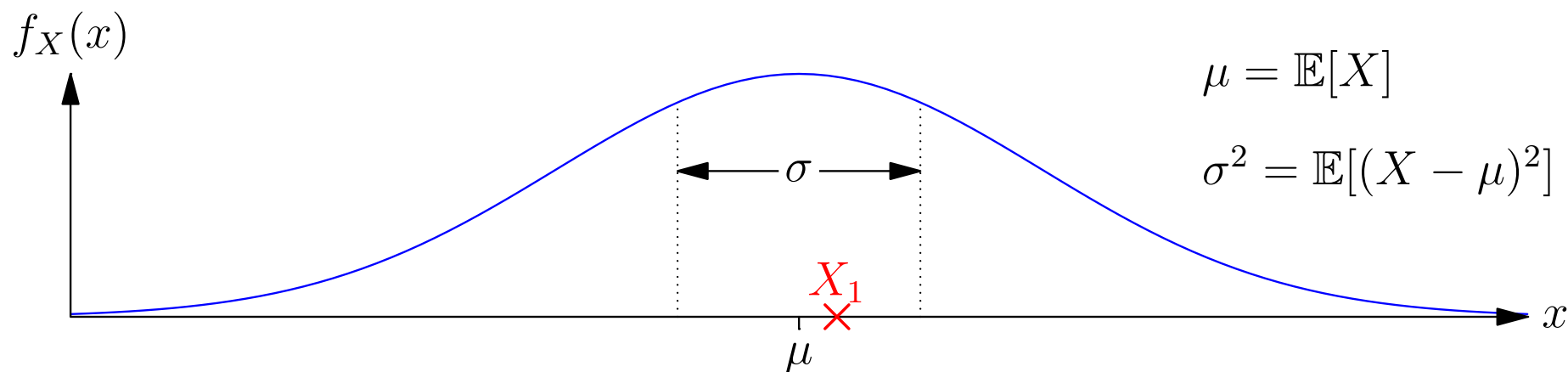
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



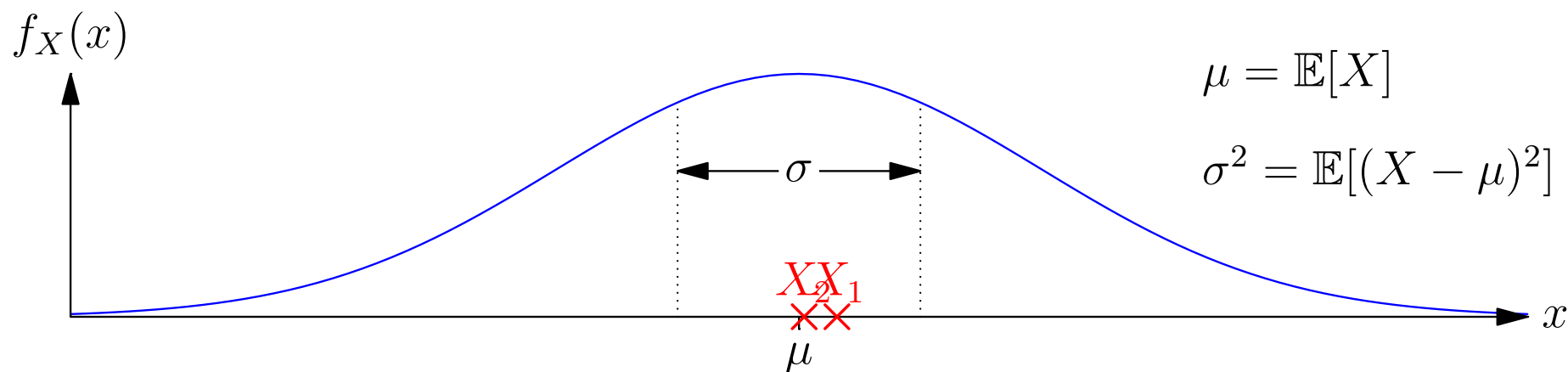
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



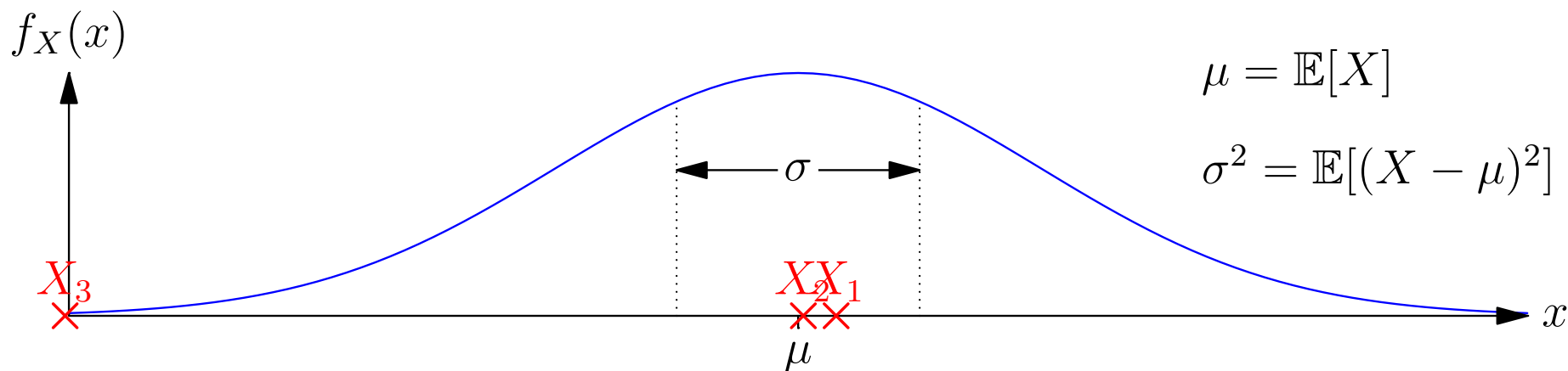
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



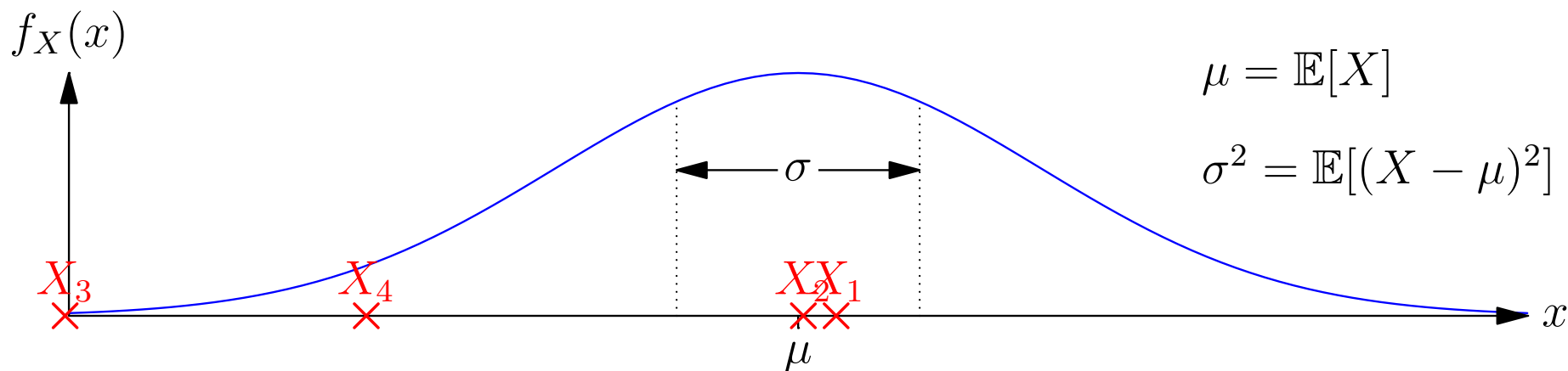
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



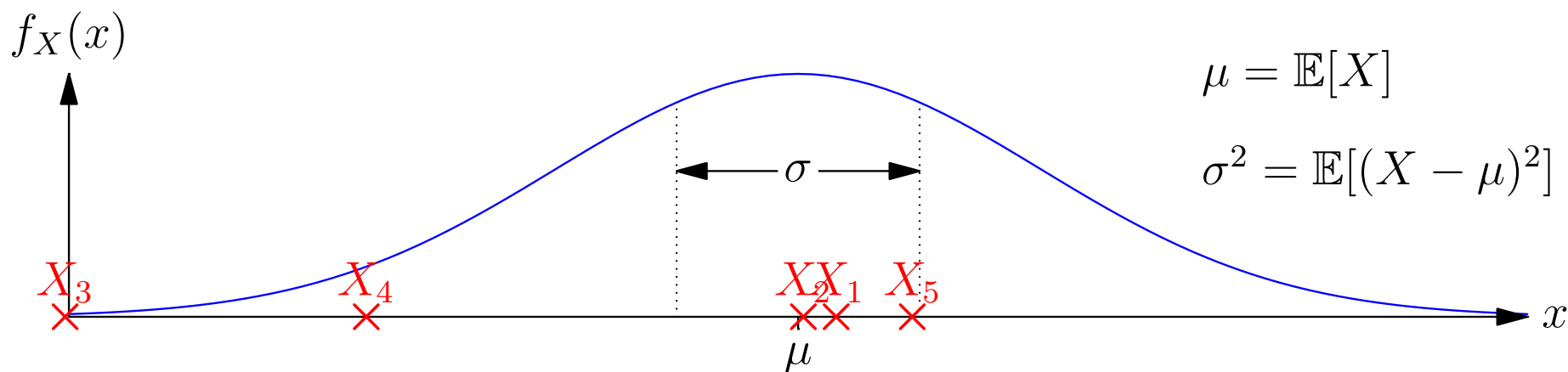
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



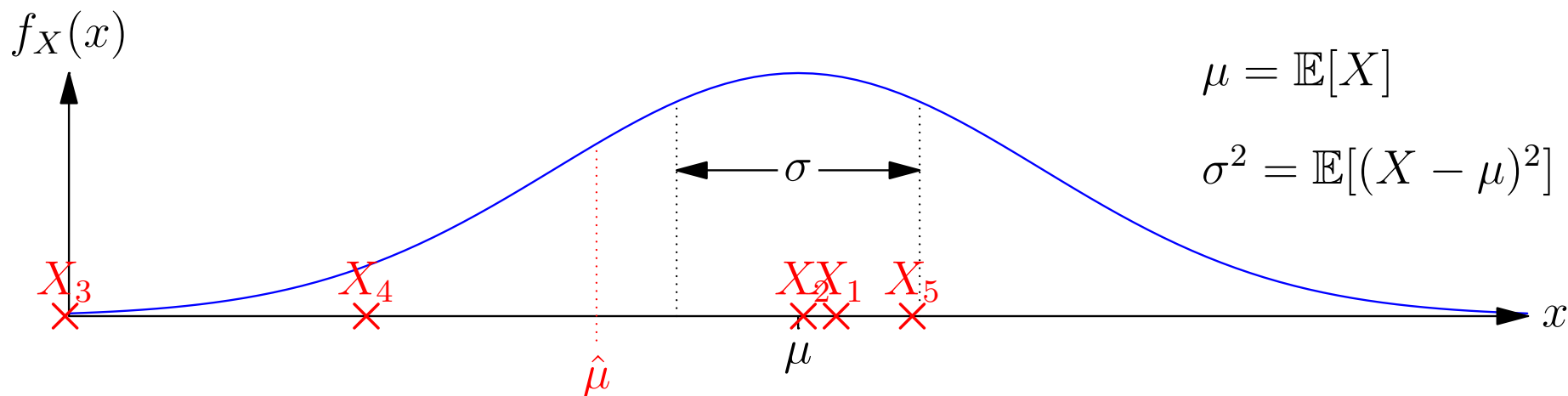
Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



Error In The Means

- By taking the mean over many samples we can reduce the variance and thus improve our generalisation performance
- To get a feel for this consider estimating the mean of a random variable, X , from a number of samples ($n = 5$ in the example below)



Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Mean and Variance

- The expected value of the mean, $\hat{\mu}_n$, of n random **independent** variables, X_i , is the expected value $\mu = \mathbb{E}[X_i]$

$$\mathbb{E}[\hat{\mu}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

- The variance is $\mathbb{E}[(\hat{\mu}_n - \mu)^2]$ or equivalently

$$\begin{aligned} \frac{1}{n^2} \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right] &= \frac{1}{n^2} \mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu] \mathbb{E}[X_j - \mu] \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{1}{n} \sigma^2 \end{aligned}$$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known a **bootstrapping**

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_1, y_1) (\mathbf{x}_2, y_2) (\mathbf{x}_3, y_3) (\mathbf{x}_4, y_4) (\mathbf{x}_5, y_5) (\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_1, y_1)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_1, y_1)$
 $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_2, y_2)$

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_1, y_1)$
 $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_2, y_2)$
 $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_1, y_1)$
 $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_2, y_2)$
 $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$
 $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_6, y_6)$

Bootstrap Aggregation (Bagging)

- To reduce the variance in a learning machine (such as a decision tree) we can average over many machines
- To average many machines they must learn something different
- We only have one data set, but we can resample from the data set to make them look a bit different—this is known as **bootstrapping**

$(\mathbf{x}_1, y_1)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$

$(\mathbf{x}_5, y_5)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_1, y_1)$
 $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_5, y_5)$ $(\mathbf{x}_2, y_2)$
 $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$
 $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_1, y_1)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_6, y_6)$
 $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_2, y_2)$ $(\mathbf{x}_3, y_3)$ $(\mathbf{x}_4, y_4)$ $(\mathbf{x}_6, y_6)$ $(\mathbf{x}_2, y_2)$

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Performance of Bagging

- Bootstrapping is an early form of data augmentation
- For classification we get our different machines to vote
- For regression we can average the prediction of different machines
- Bagging improves the performance of decision trees
- However, we can usually do better using Boosting
- This is because our decision trees are correlated

Variance of Positive Correlated Variables

- If we calculate the variance of the mean of positively correlated variables with correlation ρ we find

$$\begin{aligned}\frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n X_i - n\mu \right)^2 \right] &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[(X_i - \mu)(X_j - \mu)] \\ &= \frac{1}{n^2} (n\sigma^2 + n(n-1)\rho\sigma^2)\end{aligned}$$

$$(\rho = \mathbb{E}[(X_i - \mu)(X_j - \mu)] / \sigma^2)$$

Variance of Positive Correlated Variables

- If we calculate the variance of the mean of positively correlated variables with correlation ρ we find

$$\begin{aligned}\frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n X_i - n\mu \right)^2 \right] &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[(X_i - \mu)(X_j - \mu)] \\ &= \frac{1}{n^2} (n\sigma^2 + n(n-1)\rho\sigma^2)\end{aligned}$$

$$(\rho = \mathbb{E}[(X_i - \mu)(X_j - \mu)] / \sigma^2)$$

Variance of Positive Correlated Variables

- If we calculate the variance of the mean of positively correlated variables with correlation ρ we find

$$\begin{aligned}\frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n X_i - n\mu \right)^2 \right] &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[(X_i - \mu)(X_j - \mu)] \\ &= \frac{1}{n^2} (n\sigma^2 + n(n-1)\rho\sigma^2)\end{aligned}$$

$$(\rho = \mathbb{E}[(X_i - \mu)(X_j - \mu)] / \sigma^2)$$

Variance of Positive Correlated Variables

- If we calculate the variance of the mean of positively correlated variables with correlation ρ we find

$$\begin{aligned}\frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n X_i - n\mu \right)^2 \right] &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[(X_i - \mu)(X_j - \mu)] \\ &= \frac{1}{n^2} (n\sigma^2 + n(n-1)\rho\sigma^2)\end{aligned}$$

$$(\rho = \mathbb{E}[(X_i - \mu)(X_j - \mu)] / \sigma^2)$$

Variance of Positive Correlated Variables

- If we calculate the variance of the mean of positively correlated variables with correlation ρ we find

$$\begin{aligned}\frac{1}{n^2} \mathbb{E} \left[\left(\sum_{i=1}^n X_i - \mu \right)^2 \right] &= \frac{1}{n^2} \mathbb{E} \left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu) \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[(X_i - \mu)(X_j - \mu)] \\ &= \frac{1}{n^2} (n\sigma^2 + n(n-1)\rho\sigma^2) = \rho\sigma^2 + \frac{1-\rho}{n}\sigma^2\end{aligned}$$

$$(\rho = \mathbb{E}[(X_i - \mu)(X_j - \mu)]/\sigma^2)$$

Variance for Non-Independent Variables

- Thus,

$$\mathbb{E}[(\hat{\mu}_n - \mu)^2] = \rho\sigma^2 + \frac{1 - \rho}{n}\sigma^2$$

- If the variables X_i are independent ($\rho = 0$) we get the same result we had before (σ^2/n)
- If X_i are all identical ($\rho = 1$) we find the expected error in the mean is σ^2 i.e. no benefit of averaging
- As $n \rightarrow \infty$ the second term vanishes, but we are left with the first term
- If we want to reduce our model's variance we need our learning machines to be unbiased and decorrelated

Variance for Non-Independent Variables

- Thus,

$$\mathbb{E}[(\hat{\mu}_n - \mu)^2] = \rho\sigma^2 + \frac{1 - \rho}{n}\sigma^2$$

- If the variables X_i are independent ($\rho = 0$) we get the same result we had before (σ^2/n)
- If X_i are all identical ($\rho = 1$) we find the expected error in the mean is σ^2 i.e. no benefit of averaging
- As $n \rightarrow \infty$ the second term vanishes, but we are left with the first term
- If we want to reduce our model's variance we need our learning machines to be unbiased and decorrelated

Variance for Non-Independent Variables

- Thus,

$$\mathbb{E}[(\hat{\mu}_n - \mu)^2] = \rho\sigma^2 + \frac{1 - \rho}{n}\sigma^2$$

- If the variables X_i are independent ($\rho = 0$) we get the same result we had before (σ^2/n)
- If X_i are all identical ($\rho = 1$) we find the expected error in the mean is σ^2 **i.e. no benefit of averaging**
- As $n \rightarrow \infty$ the second term vanishes, but we are left with the first term
- If we want to reduce our model's variance we need our learning machines to be unbiased and decorrelated

Variance for Non-Independent Variables

- Thus,

$$\mathbb{E}[(\hat{\mu}_n - \mu)^2] = \rho\sigma^2 + \frac{1 - \rho}{n}\sigma^2$$

- If the variables X_i are independent ($\rho = 0$) we get the same result we had before (σ^2/n)
- If X_i are all identical ($\rho = 1$) we find the expected error in the mean is σ^2 i.e. no benefit of averaging
- As $n \rightarrow \infty$ the second term vanishes, but we are left with the first term
- If we want to reduce our model's variance we need our learning machines to be unbiased and decorrelated

Variance for Non-Independent Variables

- Thus,

$$\mathbb{E}[(\hat{\mu}_n - \mu)^2] = \rho\sigma^2 + \frac{1 - \rho}{n}\sigma^2$$

- If the variables X_i are independent ($\rho = 0$) we get the same result we had before (σ^2/n)
- If X_i are all identical ($\rho = 1$) we find the expected error in the mean is σ^2 i.e. no benefit of averaging
- As $n \rightarrow \infty$ the second term vanishes, but we are left with the first term
- If we want to reduce our model's variance we need our learning machines to be unbiased and decorrelated

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Random Forest

- In random forests we average much less correlated trees
- To do this for each tree we choose a subset of $p' \ll p$ of the features on which to split the tree
- Typically p' can range from 1 to $\sqrt{p}$
- The trees aren't that good, but are very decorrelated
- By averaging over a huge number of trees (order of 1000) we typically get good results
- Random Forest won (wins?) many competitions

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive

Lessons

- Ensemble methods have proved themselves to be very powerful
- They work by averaging over different machines, trying to reduce their variance
- Here the variance comes from forcing the machines to learn different functions using Bootstrap Aggregation
- Tend to work best with very simple models (true of random forest and boosting)—seems to reduce over-fitting
- Random forest is very powerful, but gradient boosting is competitive